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| **PI Name:** | Pluth, Janice Ph.D. | | |
| **Project Title:** | Impact of Age, Genetics Variants and High LET Track Structure on Mammary Cancer Risk Estimates | | |
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| **Task Description:** | A key NASA objective is to obtain quantitative data defining the risk of carcinogenesis following galactic cosmic ray (GCR) exposure. Current radiation quality factors used to determine cancer risk for space radiation do not provide the accuracy needed, and debate exists as to the relevance of the experimental studies they're based upon. To formulate better risk estimates, key data is needed including how age, genetic susceptibility, and radiation quality affect carcinogenic potential in irradiation sensitive tissues. Towards this aim, we will use a novel resource of a large collection of normal mammary epithelial cell strains, from women aged 16 to 80, including many in the age of astronauts, to address these important questions. We will obtain quantitative data regarding radiation quality effects; ROS (reactive oxygen species) and DNA damage signaling, radiation-induced centrosome defects, and enrichment of stem cell populations using this cohort of strains from various aged individuals. In addition, we will also investigate | | |

genetic susceptibility, including how loss of p16 expression, a sporadic event in individual cells of normal women in vivo and often observed in breast tumors, affects radiation quality effects and centrosome aberration frequency. Additionally we will analyze strains containing a BRCA1 or an ATM heterozygote genetic background, representing ~0.32 and ~1% of the normal population, respectively, to determine how these genetic variants may make them more sensitive to high LET (linear energy transfer) exposures. These studies will provide critical data, which can be used to model how genetic susceptibility, age, and radiation quality interplay in defining the cancer risk of mammary cells to high LET radiation.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

The intent of this grant was to determine how age, genetic susceptibility, and radiation quality modify the risk of high LET radiation induced cancer. A cohort of human mammary epithelial cell (HMEC) strains isolated from women of various age groups was used for these studies. Strains from individuals who are heterozygote carriers of genes involved in DNA damage response were also included for the purpose of studying genetic susceptibility and the degree of individual variability that might be expected. Centrosome studies were performed using standard microscopy-based method. As previous studies have revealed that changes in stem cell numbers are associated with an increased cancer risk, we assessed how radiation quality influenced stem cell populations and whether age played a role in the differences observed. We also assessed the phosphorylation profiles of key nodes in DNA damage signaling response and changes in miRNA profiles of a subset of strains in an attempt to tease out the mechanistic underpinnings of the influence of radiation on frequency of centrosome aberrations and changes in stem cell numbers.
Materials & Methods. Frozen vials of primary cells from eighteen individuals were obtained from an HMEC Bank at LBNL (Lawrence Berkeley National Laboratory), generated by Dr. Martha Stampfer. Early passage (p=2) cells were expanded and used for subsequent experiments. Approximately seventy p4 vials were frozen down from each individual's strain to have sufficient cell stocks to perform all experiments detailed in the grant. At BNL (Brookhaven National Laboratory) cells were exposed to three different doses (0, 0.05, and 0.5 Gy) of cesium, Si 93, and Ti 300 MeV/u in triplicate. Each strain was cultured in a T25 or a T75 flask at BNL, exposed to radiation, and then one day post exposure, cells were either sent back live to LBNL or trypsinized and frozen down for analysis after long term culture post radiation exposure. At LBNL, cells were plated at seeding densities to enable analysis at 85% confluency at different time points post radiation exposure (3, 9, and 14 days).

Summary of research accomplishments. These studies required extremely labor-intensive tissue culture, scoring for centrosome defects using microscopy, and flow cytometry determination of stem cell population changes. As a majority of studies to date typically defined genomic instability at early times post exposure (3 days) post exposure, we assessed centrosome aberrations at two later time points as well, weeks post exposure; these are time points likely more relevant and useful in modeling cancer risk. Our aim was to compare early effects at day 3 to the persistent effects of radiation exposure at longer times. We examined how centrosome levels are affected by the age of the individual the strain was derived from and the radiation quality of the exposure. We also defined the percentage of stem/progenitor cells in these various exposed populations at long times post exposure to define if radiation influences the growth of this population and how age may impact this endpoint. This data was published in the journal of Aging in early this year [Sridharan DM, Enerio S, LaBarge MA, Stampfer MM, Pluth JM. Aging (Albany NY). 2017 Feb 28;9(3):665-686. doi: 10.18632/aging.101183]. Furthermore, to gain additional information from this cohort of samples, we also performed assays using a smaller subset to assess the phosphorylation profile of select proteins involved in DNA damage response and miRNA expression from various age groups following high or low LET radiation exposure. This data is being drafted into a second manuscript that we intend to submit soon. Centrosome studies. Genetic instability was quantified based on levels of centrosome aberrations. This work sought to address whether age was an important defining characteristic in describing the effectiveness of the various radiation quality exposures to induce centrosome aberrations. These studies revealed a significant dose dependent increase in centrosome aberrations. In assessing age effects, we note that the percentage of cells with supernumerary centrosomes increases linearly with age particularly when exposed to Ti as compared to low LET. For Ti, the slopes for low dose (LD) and high dose (HD) appear parallel, with high dose exhibiting an ~2 fold increase in centrosome aberrations relative to low dose, for the same age. The baseline for centrosome aberrations in unexposed cells shows a slight increase with age that is not significant. In evaluating the distribution of the number of centrosomes in aberrant cells, our results indicate a linear increase in the proportion of cells with >3 centrosomes with both low and high doses of Ti exposure. Time course studies with assessments at day 3, 9, and 12 post exposure revealed that, in comparison to day 3 and day 9, there is a significant age dependent linear increase in the frequency of centrosome aberration at day 12. Interestingly, the fraction of cells exhibiting aberrant pATF2 foci post Ti exposure at Day12 decreased with increasing age, suggesting that the pattern of repair signaling is opposite of what we see with centrosome aberrations. In summary, this work suggests both age and radiation quality affects the levels of radiation induced-genomic instability.

Determining how radiation quality affects the enrichment of progenitor/stem cells, and how age and genetic background may modify this effect. We used Aldefluor as a flow-based marker to specifically mark and identify cells having stem/progenitor (S/P) cell potential. The three sets for each cell line were processed simultaneously, to robustly compare differences between the three independent experiments. Aliquots from the same cultured batches of samples were used for processing both centrosome aberrations and ALDH to compare endpoints within the same experiment and eliminate variability due to tissue culture.

To positively identify ALDH bright cells, cells were treated with ALDH alone or with ALDH + DEAB (inhibitor of ALDH activity) and processed immediately. Live vs dead cell discrimination was carried out by counter staining cells with DAPI. Results were analyzed using BD FACS Vantage DIVA (Becton Dickinson) and a cooled sample port. DEAB staining for each sample was used to assess the correct gating to define the percentage of ALDH positive (+) cells.

Effect of radiation exposure on stem/progenitor populations. Samples were exposed to both a low and high dose of Cs and Ti, cultured for 9 days, and then processed. Single cell suspensions were analyzed for ALDH+ cells to assess radiation-dependent changes in S/P cells. Only eleven HMEC strains were used for ALDH analysis, as we did not have sufficient numbers of viable cells for 4 strains (19, 27, 46, and 72 year old). In analyzing the entire cohort, we noted a radiation-induced increase in the mean proportion of ALDH+ cells relative to control for both types of radiation. Of interest, this increase was noted even with a low dose (5 cGy) exposure. A comparison of the effect of two different doses of Cs indicates that the median increase in ALDH+ cells is significantly greater with high dose relative to low

dose exposure. A similar although not significant trend was observed with Ti exposure (p=0.08).

Age of the cell strain impacts radiation-dependent changes in S/P population. Regression analysis was carried out to assess the relationship between age and the radiation-dependent increase in S/P cells. While there does not appear to be a correlation between the frequency of ALDH+ cells and age with exposure to low dose Cs, high dose exposure elicits a slight increase with increasing age. In contrast, when HMEC strains were exposed to complex damages from Ti exposure, the proportion of ALDH+ cells increased in an age-dependent manner at a low but not a high dose. High dose Ti exposure uniformly increased the proportion of S/P cells in all age groups.

We further compared dose effects within each age group following exposure to both types of radiation. For the most part, age-grouped analysis confirmed the trends observed with linear regression with one exception. Although there was an age-related increase in S/P numbers with high dose Cs exposure, age-grouping indicated that the mean frequency of S/P cells in the older cohort was higher than both the young and middle age groups. In addition, within each aged cohort (young, middle, and old), we observed a relative increase in the proportion of ALDH+ cells with high dose Cs exposure in comparison to low dose. A similar dose dependent increase has also been noted within each of the three age groups post Ti exposure. Thus an age and dose-dependent increase in the numbers of S/P cells is observed following radiation exposure.

Mechanistic clues that underlie age effects. Cells activate various signaling networks in response to radiation, which play a critical role in regulating various cell fates, such as cell survival, proliferation, apoptosis, senescence, repopulation, and cellular reprogramming. We hypothesized that the cellular response to the radiation would be unique to the nature of damage and show an age dependence. To test the hypothesis that the signaling response would be unique to the nature of damage and depend on age of the individual the cells were from, we assessed the global phosphorylation profile of key nodes in three different stress-activated signaling pathways, namely the receptor tyrosine kinase signaling pathway, TNF alpha signaling pathway, and TGFß signaling by multiplex analysis using the Luminex system. The fold change in phosphorylation relative to unexposed control was graphed from two independent experiments. Regression analysis revealed a pronounced age effect in a subset of these proteins that highlights differences in response to simple and complex lesions. While, low dose Cs exposure appeared to show an inverse linear relationship for phosphorylation of ERK/MAP kinase ½ (Thr185/Tyr187) with age, we noted a uniform increase in phosphorylation with high dose Cs exposure that was independent of age. The level of pAKT (Ser473), pCREB (Ser133), and total TGFßRII protein levels remain unchanged with Cs dose and did not show an association with age. In contrast, in response to complex lesions from Ti ion exposure, we noted a linear relationship between age and the relative change in the level of phosphorylation of ERK/MAP kinase ½ (Thr185/Tyr187), pAKT (Ser473), and pCREB (Ser133) following both doses. The increase was especially significant for pAKT with low dose exposure ($P<0.05$). While total protein levels of TGFßRII appear to reciprocally decrease with age at low doses of Ti exposure, we do not observe an association with age at high doses. Interestingly we observed that phosphorylation profiles also provide a unique handle to study strain specific differences in age matched samples. Our data revealed that unique individual expression signatures that change based on dose and lesion complexity.

**Task Progress:**

Mechanisms that underlie age effects of centrosome aberrations. To test whether the increase in centrosome aberrations was unique to Ti300 exposure, we exposed a subset of the aged cohort of cell strains to Si 93 MeV/n and assessed centrosome aberrations 9 days post exposure. Our results revealed that the lower energy ion, Si 93 MeV/n, was more effective in generating centrosome aberrations in comparison to Ti 300 MeV/n, and showed a similar age dependent increase with a steeper slope than Ti 300 MeV/n. Inflammation is known to play a key role in the cellular response to radiation. As microRNA (miRs) are increasingly thought to influence cellular response by targeted repression of the expression of proteins involved in various cell fate choices, we examined the microRNA signature of a subset of strains (2) from the young, middle, and old age groups that were exposed to Si 93 MeV/n. We specifically tested a panel of 68 miRs that have been shown to be important in inflammation, to define whether age-specific patterns in miRs are evident.

miRNA expression was blank subtracted and normalized using 3 internal normalizers (let-7i-5p, miR 23a-3p, and miR 30b-5p) based on their expression pattern. Only miRs within the limits of detection as identified by the software were included for analysis.

Of the 68 miRs tested, we identified a subset of miRs where the expression levels decrease with age (10) and a second subset where the expression level increases with age (6). While some of the identified age dependent miRs expression patterns are observed both at low and high doses (7d-5p, 7e-5p, 7g-5p, 20b-5p, 29c5p, 93-5p, 15a-5p, 20a-5p, 320d, and 130a-5p) others showed an age dependent expression pattern that is unique to dose (155-5p,146a-5p, 24-3p, 130a-5p, 24-3p, and 181a-5p). Of these miR-320d shows a significant linear decrease in expression with age with high dose Si 93 MeV/n exposure, miR-24-3p shows a significant linear increase with age with both low and high dose Ti exposure, and miR- 29b-3p shows a significant increase that is unique to high dose exposure. These results are promising, as a number of miRs that we identified have been linked to carcinogensis. For example, we observe miR29b to show an increase with dose and aberrant expression of miR-29b that has been shown to contribute to migration, invasion, and anti-apoptosis. In addition, a dose-dependent increase in miR-146a was shown in our work and an increase in levels has also been observed in breast cancer cells.

For statistical analysis, the expression level of each miR target in each sample (MFI) was determined and sorted by appropriate groups. For each group, we calculated the mean signal within the groups, and ANOVA statistics are used to calculate p-values. One way anova of miR expression level versus radiation exposure revealed that although 1 miR target (192-5p) showed a significant difference between the radiation doses (raw $p<0.05$), correcting for multiple comparisons using Boferroni's test revealed that miR-192-5p was not significantly different between exposed and sham samples. One way anova of miR expression level versus age revealed three miRNA targets that showed significant differences between age groups even after correcting for multiple comparisons. These are: miR146a-5p, miR 15b-5p, and miR 93-5p ($p<0.05$). Box and whisker plots were used to present differential expression analyses. Two way ANOVA vs age controlled for radiation exposure revealed that controlling for radiation did not show a significance difference between age groups, suggesting that radiation had very little effect and confirming the results of one way ANOVA analysis.

When each individual was plotted separately, we can clearly identify strain specific effects. While the majority of the strains show no differences between the different doses, the 65 year old shows a clear change in expression of several of the miRNAs, suggesting perhaps that this individual is an outlier in terms of radiation response.

We next, selected the miRs that were most differentially expressed with age, and created a heat map controlled for radiation, and clustered probes by similarity. Analyzed in this manner we note some general trends with specific miRs that seem to show higher expression with radiation exposure. A heat map also identifies that cells from one individual

has an expression pattern that was an outlier as compared to other strains.

Genomic instability in heterozygote carriers of genes involved in repair response. In addition to age, genetic variation is a key parameter that modifies cancer risk in response to radiation exposure. The increase in cancer susceptibility of heterozygote carriers for DNA repair genes like BRCA1, BRCA2, and AT is highly debated. While several studies suggest that heterozygote carriers are at an increased risk for carcinogenesis, whether this is mimicked at the cellular level, by an increase in genomic instability in response to radiation exposure is unclear. To answer this question, we utilized a select cohort of well-characterized human mammary epithelial strains from various genetic backgrounds (BRCA1+/-, BRCA2+/-, AT+/1, P16+/-) to directly address how individual genetics influence the development of cellular surrogate markers of cancer risk. Cells have built in mechanisms such as apoptosis and stress-associated senescence, also termed stasis, that serve as barriers to transformation. Thus, evasion of stasis is one of the first steps towards transformation. Loss of p16 can overcome the stasis barrier and has been widely observed in mammary tumors. Interestingly, spontaneous loss of p16 has also been noted within the normal cell population. We have compared a normal strain from a 21 y old, with and without loss of p16 using RNAi, to evaluate genomic instability post radiation exposure in cells with loss of p16 expression. The normal and the heterozygote HMEC strains were exposed to a low or high dose of Cs and Si93 MeV/n. Centrosome aberrations were evaluated at day 3 and 9 post exposure.

We noted that heterozygote carriers had higher frequencies of centrosome aberrations as compared to a representative normal strain. Following high dose exposure the centrosome aberration frequency per unit Gy was showed a greater increase with Si93 MeV/n exposure as compared to Cs. Following a high dose exposure the centrosome aberration levels in the BRCA1+/- strain were much higher than the BRCA2+/- carrier. At low doses where ~25% of the cells receive a particle traversal, BRCA1+/- cells continue to show a higher centrosome aberration frequency with Si93 MeV/n exposure. These results suggest that these carriers are especially sensitive to the complex lesions caused by high LET exposure. As BRCA1 has a significant role in centrosome maintenance, BRCA1 carriers might be more predisposed to the generation of centrosome aberrations as compared to BRCA2+/- and AT+/- heterozygote strains.

To further understand if radiation induced signaling would provide clues to the differences in sensitivity, we carried out multiplexed phosphorylation profiles and expression profiles of key proteins involved in DNA damage response at 1 and 3 days post exposure. Targets analyzed include gammaH2AX, pChk1, pChk2, pP53, and total levels of MDM2, ATR, and P21. The fold change per unit dose normalized to control was plotted for each of the heterozygote carriers in comparison to control. The results indicate that while expression levels change between the heterozygote carriers and control strain post cesium exposure, the expression of all of the protein targets were higher than the control at 72 h post Si93 MeV/n exposure. The pattern at 24 h was very similar to that at 72 h.

For BRCA1 and BRCA2 carriers, the biggest difference in expression as compared to control was noted at 24 h post Si93 exposure. There appears to be a direct correlation with the pattern of increase in protein expression relative to control and the increase in centrosome aberrations, suggesting that processes such as cell cycle arrest induced by these signaling nodes are key in generating increased centrosome aberrations in heterozygote carriers.

Overall summary. In total our studies thus far have yielded robust results that would be invaluable in modeling long-term effects of high LET exposure. The etiology of breast cancer involves a complex interplay amongst various factors such as repair capability, oxidative stress, genomic instability, stem cell effects, and integrity of tissue architecture that contribute to susceptibility in all age groups. In light of our results, we postulate that when HMECs from older individuals are exposed to complex damages, this exposure initiates a cascade of interconnected events.

Poor repair capacity in the older cohort impairs repair fidelity. In older cells, impaired redox balance caused by complex damages may further hamper lesion repair by enhancing protein oxidation and nuclear disorganization. This results in cells with unrepaired and mis-repaired lesions that cause cell cycle arrest. In a fraction of cells with either prolonged cell cycle arrest or fragile centrosomes or both, the centrosome cycle is deregulated resulting in the generation of supernumerary centrosomes. As oxidative stress has also been shown to contribute to the generation of centrosome aberrations, the pro-oxidant state in older people is likely to enhance CA frequency and further potentiate genomic instability. At the cellular level, centrosome amplification can result in the formation of multipolar spindles that can slow down mitosis and cause mitotic abnormalities. Cells have inherent mechanisms to protect against transformation by inducing mitotic cell death, apoptosis, or senescence. However, when these genome-destabilizing events occur in stem cells, they could cause catastrophic results by increasing proliferation of this unstable stem cell population while simultaneously allowing them to evade apoptosis, both of which are intrinsic features in carcinogenesis. Although we don’t observe a linear association between radiation-induced increase in stem cells and age, we do see an age effect when comparing grouped ages. Pearson’s correlation analysis has also revealed a strong association between ALDH+ cells and centrosome aberration frequency. This higher centrosome aberration frequency in cells from older individuals could increase the potential for genomic instability within the stem cell population. This increased mutational load will likely impact their ability to differentiate into various cell types thus altering composition and impairing the structural integrity of the tissue. The results of these studies will be crucial to quantitatively factor in the influence of age and radiation quality on genomic instability and stem cell populations, endpoints that have been linked to cancer. Therefore these studies should provide data useful for modeling the risk of mammary cancer from space radiation exposures and decrease the uncertainties due to age and genetic variation.

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| **Bibliography Type:** | Description: (Last Updated: 11/01/2023) |
| **Articles in Peer-reviewed Journals** | Sridharan DM, Enerio S, LaBarge MA, Stampfer MM, Pluth JM. "Lesion complexity drives age related cancer susceptibility in human mammary epithelial cells." Aging (Albany NY). 2017 Feb 28;9(3):665-86. http://dx.doi.org/10.18632/aging.101183 ; PubMed PMID: 28245431; PubMed Central PMCID: PMC5391225 , Feb-2017 |
| **Articles in Peer-reviewed Journals** | Sridharan DM, Asaithamby A, Blattnig SR, Costes SV, Doetsch PW, Dynan WS, Hahnfeldt P, Hlatky L, Kidane Y, Kronenberg A, Naidu MD, Peterson LE, Plante I, Ponomarev AL, Saha J, Snijders AM, Srinivasan K, Tang J, Werner E, Pluth JM. "Evaluating biomarkers to model cancer risk post cosmic ray exposure." Life Sci Space Res (Amst). 2016 Jun;9:19-47. Review. http://dx.doi.org/10.1016/j.lssr.2016.05.004 ; PubMed PMID: 27345199 , Jun-2016 |

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| **Articles in Peer-reviewed Journals** | Sridharan DM, Enerio S, Wang C, LaBarge MA, Stampfer MR, Pluth JM. "Genetic variation and radiation quality impact cancer promoting cellular phenotypes in response to HZE exposure." Life Sci Space Res (Amst). 2019 Feb;20:101-12. Epub 2018 Oct 21. https://doi.org/10.1016/j.lssr.2018.10.002 ; PubMed PMID: 30797427 , Feb-2019 |